Chapter 19

Chemical Thermodynamics

1. The first law of thermodynamics can be given as ____A____.

A) $\Delta E = q + w$
B) $\Delta H^0_{rxn} = \sum n\Delta H^0_f\,(products) - \sum m\Delta H^0_f\,(reactants)$
C) for any spontaneous process, the entropy of the universe increases
D) the entropy of a pure crystalline substance at absolute zero is zero
E) $\Delta S = q_{rev}/T$ at constant temperature

The first law of thermodynamics: energy cannot be created or destroyed, but can be converted from one to another.

$$\mathbf{\Delta E = q + w}$$

2. Which one of the following is always positive when a spontaneous process occurs? ___C___

A) ΔS_{system}
B) $\Delta S_{surroundings}$
C) $\Delta S_{universe}$
D) $\Delta H_{universe}$
E) $\Delta H_{surroundings}$

3. ΔS is positive for the reaction ____C____.

A) $2NO\ (g) + O_2\ (g) \rightarrow 2NO_2\ (g)$
B) $2N_2\ (g) + 3H_2\ (g) \rightarrow 2NH_3\ (g)$
C) $C_3H_8\ (g) + 5\ O_2\ (g) \rightarrow 3CO_2\ (g) + 4\ H_2O\ (g)$
D) $Mg\ (s) + Cl_2\ (g) \rightarrow MgCl_2\ (s)$
E) $C_2H_4\ (g) + H_2\ (g) \rightarrow C_2H_6\ (g)$

4. The standard Gibbs free energy of formation of _D_ is zero.

(a) H_2O (l)

(b) Fe (s)

(c) I_2 (s)

Gas
Liquid } 25°C, 1 atm

A) (a) only

B) (b) only

C) (c) only

D) (b) and (c)

E) (a), (b), and (c)

Element	$\Delta G_f^\circ = 0$ for element in standard state

5. For the reaction

$$2\ C_4H_{10}\ (g)\ +\ 13\ O_2\ (g) \rightarrow\ 8\ CO_2\ (g)\ +\ 10\ H_2O\ (g)$$

$\Delta H°$ is -125 kJ/mol and $\Delta S°$ is +253 J/K · mol. This reaction is ___A___.

$\Delta G° < 0 \quad \forall T$

A) spontaneous at all temperatures
B) spontaneous only at high temperature
C) spontaneous only at low temperature
D) nonspontaneous at all temperatures
E) unable to determine without more information

$$\Delta G = \Delta H - T\Delta S$$

6. Given the following table of thermodynamic data,

Substance	ΔH_f° (kJ/mol)	S° (J/mol • K)
$TiCl_4$ (g)	–763.2	354.9
$TiCl_4$ (l)	–804.2	221.9

$\Delta H > 0$ $\Delta S > 0$

complete the following sentence. The vaporization of $TiCl_4$ is ___C___.

$TiCl_{4(l)} \rightarrow TiCl_{4(g)}$ $\Delta G^\circ < 0$

A) spontaneous at all temperatures
B) spontaneous at low temperature and nonspontaneous at high temperature
C) nonspontaneous at low temperature and spontaneous at high temperature
D) nonspontaneous at all temperatures
E) not enough information given to draw a conclusion

7. With thermodynamics, one cannot determine ___A___.

A) the speed of a reaction kinetics (k)
B) the direction of a spontaneous reaction ΔG
C) the extent of a reaction
D) the value of the equilibrium constant $\Delta G = RT\ln K$
E) the temperature at which a reaction will be spontaneous $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

8. Consider the reaction:

$$FeO\,(s) + Fe\,(s) + O_2(g) \rightarrow Fe_2O_3\,(s)$$

Given the following table of thermodynamic data at 298 K:

The value K for the reaction at 25 °C is ___D___.

A) 370

B) 5.9×10^{4}

C) 3.8×10^{-14}

D) 7.1×10^{85}

E) 8.1×10^{19}

	Substance	ΔH_f° (kJ/mol)	S° (J/K • mol)
−290 −8.09	FeO (s)	-271.9	60.75
	Fe (s)	0	27.15
−61.09	O_2 (g)	0	205.0
−848.97	Fe_2O_3 (s)	-822.16	89.96

-489.79 kJ/mol $= 8.31$ J/mol·K $\cdot$ 298 K $\cdot \ln K$

$K = 7.1 \times 10^{85}$

$\log_{10} e^{197.78} =$

$197.78 \times \log_{10} e$

9. The normal boiling point of methanol is 64.7 °C and the molar enthalpy of vaporization is 71.8 kJ/mol. The value of ΔS when 1.75 mol of CH_3OH (l) vaporizes at 64.7 °C is ___C___ J/K.

$\Delta G^\circ = 0$ 337.85

$\rightarrow \Delta S^\circ = \frac{\Delta H^\circ}{T} = \frac{71.8\ kJ/mol \cdot 1.75\ mol}{337.85\ K}$

A) 0.372
B) 4.24×10^7
C) 372
D) 1.94×10^3
E) 1.94